# Black Holes: A Primer

## 1. What Is a Black Hole?

A black hole is a region of spacetime where gravity is so intense that nothing, not even light, can escape. The boundary of no return is the **event horizon**.

The idea dates back to the 18th century (John Michell and Pierre-Simon Laplace speculated about 'dark stars'). The modern description comes from Einstein's general relativity (1915) and Karl Schwarzschild's 1916 solution.

Key properties are fully described by just three quantities: mass, spin, and electric charge (the 'no-hair theorem').

# 2. Formation and Types

**Stellar-mass black holes** form when massive stars collapse at the end of their lives, typically producing 3 to 100 solar masses.

**Supermassive black holes** ($10^6$ to $10^{10}$ solar masses) reside at galactic centers. The Milky Way hosts Sagittarius A* at about 4 million solar masses.

**Intermediate-mass black holes** are a missing link with few confirmed detections. **Primordial black holes** are hypothesized to have formed shortly after the Big Bang.

The Schwarzschild radius is $r_s = 2GM/c^2$. For the Sun, this would be about 3 km.

# 3. Observing Black Holes

Because black holes emit no light, we observe their effects: stellar orbits near Sagittarius A*, X-ray emission from hot accretion disks, and relativistic jets.

In 2015, LIGO made the first direct detection of gravitational waves from a black hole merger (GW150914). In 2019 the Event Horizon Telescope released the first image of a black hole shadow (M87*).

Hawking radiation, predicted in 1974, suggests black holes slowly evaporate over immense timescales. It has not been observed.